

VOI, Kenya

WMO: 637930

Lat: 3.398S

Lon: 38.558E

Elev: 560

StdP: 94.78

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.1	18.8	11.3	8.9	27.4	12.2	9.5	27.1	13.8	25.0	12.5	25.6	4.5	270	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	10.9	34.6	22.2	33.8	22.2	32.9	22.1	24.5	29.7	24.1	29.3	23.8	28.8	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.3	25.7	23.0	19.0	25.5	22.6	18.6	25.3	77.4	29.5	75.8	29.5	74.4	29.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
12.1	10.9	9.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			14.2	26.0	1.5	1.5	13.1	27.1	12.2	28.0	11.4	28.8	10.4	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	26.6	27.2	27.8	27.1	25.4	24.6	23.7	23.6	24.3	25.7	26.3	26.3
	DBStd	1.75	1.01	1.22	1.11	1.26	1.17	1.03	0.93	1.05	1.09	1.24	1.19	1.01
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5730	514	483	552	512	479	437	424	420	430	486	488	504
	CDD18.3	2688	256	249	294	262	221	187	166	162	180	228	238	246
	CDH23.3	20451	2173	2322	2813	2202	1483	1125	891	853	1174	1683	1833	1898
	CDH26.7	7449	887	1067	1309	866	438	277	163	159	335	626	674	647
Wind	WSAvg	4.5	3.3	3.2	3.6	4.0	5.5	5.6	6.2	6.0	5.5	4.5	3.5	2.9
Precipitation	PrecAvg	577	35	22	91	88	27	4	3	6	15	30	132	116
	PrecMax	941	144	104	385	274	160	32	14	32	93	122	334	259
	PrecMin	280	0	0	8	3	1	0	0	0	0	0	4	18
	PrecStd	179	42	23	87	58	30	7	4	8	24	31	80	72
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	35.6	35.7	34.9	32.4	31.2	30.4	31.0	32.2	33.2	33.8	32.7
		MCWB	21.7	22.0	22.7	22.7	21.8	20.1	19.5	20.0	20.2	21.4	22.5	22.6
	2%	DB	33.1	34.4	34.6	33.4	31.1	30.1	29.4	29.4	30.9	32.2	32.3	31.8
		MCWB	21.8	21.9	22.4	22.9	21.6	19.7	19.3	19.6	20.0	21.2	22.2	22.5
	5%	DB	32.1	33.4	33.7	32.2	30.1	29.2	28.5	28.3	29.8	31.2	31.3	30.9
		MCWB	21.9	21.9	22.4	22.9	21.4	19.7	19.1	19.3	19.8	21.1	22.2	22.5
	10%	DB	31.0	32.2	32.7	31.1	29.2	28.4	27.5	27.4	28.8	30.1	30.3	30.1
		MCWB	22.0	21.9	22.5	22.9	21.3	19.7	18.9	19.2	19.7	20.9	22.2	22.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	24.7	25.0	24.8	23.7	21.9	21.2	21.6	22.8	23.7	24.2	24.7
		MCDB	29.5	29.7	31.3	30.1	28.7	27.0	27.5	25.9	28.3	28.1	28.4	29.0
	2%	WB	23.7	23.9	24.4	24.2	23.0	21.4	20.5	20.9	21.6	22.9	23.8	24.2
		MCDB	28.8	29.3	30.4	29.7	27.9	26.3	26.4	25.9	27.1	27.6	28.0	28.5
	5%	WB	23.3	23.4	24.0	23.7	22.6	21.0	20.0	20.4	21.0	22.4	23.5	23.8
		MCDB	28.4	28.8	29.8	29.1	27.3	25.9	25.7	25.6	26.4	27.3	27.7	27.9
	10%	WB	22.9	23.0	23.6	23.5	22.1	20.6	19.6	19.9	20.4	22.0	23.1	23.4
		MCDB	27.9	28.4	29.1	28.5	26.7	25.5	25.1	25.3	25.9	26.9	27.3	27.5
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.8	10.9	9.3	8.3	8.7	8.9	8.7	9.7	9.9	9.2	8.5
		MCDBR	11.3	12.5	12.1	10.6	9.6	10.0	10.1	10.1	11.2	11.5	10.6	9.4
	5% WB	MCWBR	2.7	2.8	2.5	2.3	2.4	2.4	2.5	2.5	2.8	2.7	2.4	2.4
		MCWBR	9.6	10.4	10.5	9.4	8.0	8.4	8.9	8.7	9.5	9.6	8.8	8.5
Clear-Sky Solar Irradiance	taub	taub	0.437	0.453	0.420	0.369	0.364	0.395	0.405	0.391	0.381	0.382	0.390	0.424
		taud	2.312	2.254	2.377	2.560	2.558	2.426	2.362	2.391	2.416	2.436	2.452	2.375
	Ebn at Noon	Ebn at Noon	900	886	905	929	908	864	861	895	927	940	937	907
		Edh at Noon	137	146	128	103	99	110	119	120	121	120	118	127
All-Sky Solar Radiation	RadAvg	RadAvg	6.12	6.31	6.17	5.50	4.87	4.56	4.54	4.85	5.57	5.95	5.87	5.94
	RadStd	RadStd	0.32	0.29	0.36	0.27	0.24	0.19	0.27	0.27	0.28	0.28	0.30	0.25

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.36	+0.47	+0.46	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page